QuantTerminal – System Documentation

QuantTerminal is a Python and Streamlit research platform for Indian (NSE/BSE) and US equities. It combines market data, statistics, forecasting, risk simulation, strategy testing, portfolio construction and PDF report generation in one app.

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Project Information

Item	Detail
Project	QuantTerminal

Item	Detail
Description	Quantitative finance research platform for Indian and US equities
Live application	quantterminal.streamlit.app
Repository	github.com/MichaelF102/Quant_Terminal
Platform	Python 3.10 to 3.12, Streamlit

Project Team

Name	Role	Responsibilities	Assigned pages
Pramit Datta	Project Lead	Strategic governance, quantitative methodology, deliverable review, roadmap and architectural validation	All workstreams
Vanshika Soni	Project Lead	Program management, workstream synchronisation, timeline governance, quality assurance and report compliance	All workstreams
Michael Fernandes	Lead Architect and Integration Owner (Engine and Advanced ML)	Core execution engine, data ingestion and caching, backtesting, forecasting, machine learning, regime detection, Monte Carlo simulation, RL environment	Backtesting, Monte Carlo Simulations, Time Series Forecasting, ML Forecasting, DL Forecasting, Regime Detection, RL Trading

Name	Role	Responsibilities	Assigned pages
Pranav Sahai	Statistics, Risk and Econometric Research	Distribution statistics, volatility estimators and GARCH, VaR and CVaR, factor research, statistical arbitrage	Return Analytics, Statistical Analysis, Volatility Lab, Risk Analytics, Factor Research, Statistical Arbitrage
Aashima Grover	Technicals, Visual Analytics, Portfolio and Reporting	Technical indicators, shared charting layer, portfolio optimisation, market explorer, strategy lab, 20-page PDF report	Dashboard, Market Explorer, Technical Analysis, Strategy Lab, Portfolio Lab, Report Generation

QuantTerminal v2.4 is designed and developed by Fincell Quants. A twentieth page, Credits, presents the team, workstream split and build timeline.

1. Overview

Purpose

The platform is a single workspace for quantitative research on listed equities. A user picks a ticker, loads daily OHLCV data, and moves through analysis, modelling, simulation and reporting without leaving the app. It is built for education and research and does not give investment advice.

Design principles

- **Point-in-time data:** analysis is built around daily OHLCV bars (the sidebar also offers weekly, monthly and intraday intervals), with a stated zero look-ahead-bias protocol.
- **No fabricated metrics:** modules that have not been run do not show invented numbers. The PDF report renders architectural flowcharts and constraint matrices in their place.
- **Market-aware hygiene:** non-trading calendar gaps are removed so they do not create false zero-return autocorrelation.
- **Reproducible output:** the same live session data feeds every page and the PDF report.

Technology stack

Layer	Technology
Language	Python 3.10, 3.11 or 3.12
Front end	Streamlit multipage app with a dark theme
Report engine	ReportLab with the DejaVu Sans TrueType font
Data source	Yahoo Finance daily OHLCV
Testing	pytest
Licence	MIT

Analytical domains

The app is organised into 7 research domains. Each domain is one or more Streamlit pages under `pages/`; later sections document them in this order.

#	Domain	Pages
1	Market & Data	Market Explorer, Technical Analysis, Dashboard
2	Statistics & Returns	Return Analytics, Statistical Analysis, Volatility Lab
3	Forecasting	Time Series, ML, DL forecasting
4	Simulation & Risk	Monte Carlo, Risk Analytics, Regime Detection
5	Strategy & Trading	Strategy Lab, Backtesting, RL Trading
6	Portfolio & Research	Portfolio Lab, Factor Research, Statistical Arbitrage
7	Research Reporting	Report Generation Center

Repository layout

Path	Contents
`app.py`	Home dashboard, the module registry of 19 entries, and thread-limit settings for cloud hosts

Path	Contents
`config.py`	Exchange options, benchmarks, currency symbols, Indian presets, cap thresholds, default risk-free rate, trading days
`pages/`	19 Streamlit pages numbered 01 to 19, listed in sections 3 to 9
`utils/`	`helper.py` (theme, data loading, hygiene, formatting) and `sidebar.py` (shared controls)
`reporting/`	`pdf_generator.py`, `report_styles.py`, `report_charts.py`, `report_data.py`
`core/`	`metrics.py` and `drawdown.py` (shared performance metrics)
`data/snapshots/`	`India_Stocks_Data.csv` and `US_Stocks_Data.csv` stock universes
`tests/`	`test_metrics.py` and `test_reporting_pipeline.py`
`.streamlit/`	`config.toml` (dark theme, headless mode)
`docs/` and `QuantTerminal_System_Documentation.pdf`	Existing project documentation
Domain packages	`agent`, `backtesting`, `factor`, `forecasting`, `machine_learning`, `ml_dl`, `optimization`, `portfolio`, `regime`, `risk`, `services`, `simulation`, `statarb`, `statistics`, `strategies`, `technical`, `volatility`, plus `plots`, `reports` and `scripts`

The 19 page files are `01_Dashboard`, `02_Market_Explorer`, `03_Technical_Analysis`, `04_Statistical_Analysis`, `05_Volatility_Lab`, `06_Regime_Detection`, `07_Factor_Research`, `08_Strategy_Lab`, `09_Statistical_Arbitrage`,

`10_Backtesting`, `11_Portfolio_Lab`, `12_Return_Analytics`, `13_Risk_Analytics`, `14_Monte_Carlo_Simulations`, `15_Time_Series_Forecasting`, `16_ML_Forecasting`, `17_DL_Forecasting`, `18_RL_Trading` and `19_Report_Generation`.

2. Getting Started

The app runs locally with Python 3.10 to 3.12 and four commands, and deploys to Streamlit Community Cloud or Docker with no extra configuration.

Local setup

```
git clone https://github.com/MichaelF102/Quant_Terminal.git
cd Quant_Terminal
python3 -m venv .venv
source .venv/bin/activate        # Windows: .venv\Scripts\activate
pip install --upgrade pip
pip install -r requirements.txt
streamlit run app.py
```

The app opens at `http://localhost:8501`.

Dependencies

All packages are listed in `requirements.txt`, grouped by role. Versions are minimums unless noted.

Group	Packages
Web framework	streamlit 1.32 or later
Data ingestion	yfinance, pandas, numpy (below 2.0), pyarrow, requests, beautifulsoup4, tradingview-screener
Visualisation	plotly, matplotlib, seaborn, kaleido, Pillow
Statistics and econometrics	statsmodels, scipy, arch, stochastic, ta
Machine learning	scikit-learn, hmmlearn, ruptures, xgboost, lightgbm, catboost
Optimisation	cvxpy, PyPortfolioOpt, numba, joblib
Deep learning	tensorflow-cpu, keras 3 or later

Group	Packages
Reinforcement learning	tensortrade, gymnasium
Documents and export	openpyxl, reportlab, pypdf, pymupdf
Testing	pytest

Runtime behaviour set by `app.py`

Before importing any numerical library, `app.py` sets the thread-count variables `OMP_NUM_THREADS`, `OPENBLAS_NUM_THREADS`, `MKL_NUM_THREADS`, `VECLIB_MAXIMUM_THREADS`, `NUMEXPR_NUM_THREADS`, `TF_NUM_INTRAOP_THREADS` and `TF_NUM_INTEROP_THREADS` to 1 with `setdefault`. It also sets `TF_CPP_MIN_LOG_LEVEL` to 2. This stops worker-thread explosions on shared cloud containers, and an existing value in the environment still takes precedence.

It then adds the project root and `utils/` to `sys.path`, so pages can import `utils.*` and top-level modules without installation.

Streamlit Community Cloud

1. Push or fork the repository to GitHub.
2. In Streamlit Cloud, create a new app that points to the repository.
3. Set the main file path to `app.py`.
4. Deploy. The `.streamlit/config.toml` file sets dark mode and headless operation.

Docker

The README supplies a Dockerfile based on `python:3.12-slim`. It installs `fonts-dejavu-core`, `build-essential` and `curl`, installs the requirements, exposes port 8501 and defines a health check against `/_stcore/health`.

```
docker build -t quantterminal:latest .
docker run -p 8501:8501 quantterminal:latest
```

The Dockerfile is not committed to the repository root; create it from the README before building.

3. Market & Data

Three pages cover price data, charting and indicators. They are the usual starting point, since every other domain reads the same ticker and date range chosen in the sidebar.

Module	Page file	Role
Dashboard	`pages/01_Dashboard.py`	Single-screen equity monitor
Market Explorer	`pages/02_Market_Explorer.py`	Charting and universe screener (featured workspace)
Technical Analysis	`pages/03_Technical_Analysis.py`	Indicator workstation

Dashboard

A consolidated equity research monitor for the selected ticker.

- Live ticker metrics and trading volume.
- Key valuation and fundamental ratios.
- Summary of technical momentum indicators.
- Return distribution across several timeframes.
- README adds trading-volume anomaly flags and cross-asset context.

Market Explorer

Interactive market data analysis and the platform's screener.

- OHLCV candlestick chart with a volume panel.
- Moving-average overlays: SMA 20, SMA 50 and SMA 200.
- Multi-cap screener across Large, Mid, Small and Micro segments, using the thresholds in section 10.
- Sector, industry and exchange breakdowns, with feeds for several exchanges.

Technical Analysis

An algorithmic indicator workstation covering momentum, volatility and trend.

- Bollinger Bands with bandwidth dynamics.
- RSI and the Stochastic oscillator.

- MACD signal line and histogram crossovers.
- Average True Range (ATR) and volatility channels.
- Moving-average ribbons (README).

Capability coverage

The module registry in `app.py` tags each page with the capabilities it provides.

Module	Analysis	Modelling	Risk	Visualisation	Export
Dashboard	Yes	No	Yes	Yes	Yes
Market Explorer	Yes	No	Yes	Yes	Yes
Technical Analysis	Yes	Yes	Yes	Yes	Yes

4. Statistics & Returns

Three pages describe how the selected asset's returns are distributed, how they compound, and how volatile they are. They feed the risk, forecasting and reporting modules.

Module	Page file	Role
Return Analytics	`pages/12_Return_Analytics.py`	Return transformations and compounding
Statistical Analysis	`pages/04_Statistical_Analysis.py`	Distribution moments and hypothesis tests
Volatility Lab	`pages/05_Volatility_Lab.py`	Volatility estimators and GARCH models

The return convention used in the PDF report is continuous log compounding, annualised with 252 trading days.

```
R_t = \ln\left(\frac{P_t}{P_{t-1}}\right), \qquad \sigma_{annual} = \sigma_{daily}\sqrt{252}
```

Return Analytics

- Continuous logarithmic returns and discrete arithmetic returns.

- Cumulative growth and an equity wealth index.
- Calendar return heatmaps, monthly and yearly.
- Rolling 30-day and 90-day return and volatility profiles.

The report summarises this module as sample-period CAGR (geometric), arithmetic annualised mean return (mean daily return times 252), annualised standard deviation, best and worst single day, and peak-to-trough drawdown.

Statistical Analysis

- Higher-order moments: skewness and excess kurtosis.
- Normality tests: Jarque-Bera and Shapiro-Wilk, each reported with its test statistic and p-value.
- Empirical KDE against a Gaussian fit, plus a normal Q-Q plot.
- Pearson, Spearman and Kendall correlation matrices across assets.

Volatility Lab

- Five realised-volatility estimators: Close-to-Close, Parkinson, Garman-Klass, Rogers-Satchell and Yang-Zhang.
- GARCH(1,1) and EGARCH conditional volatility models.
- Rolling percentile volatility cones from 10 to 252 days.
- Annualised volatility term structure and clustering views.

The PDF report adds an EWMA estimator (RiskMetrics, lambda 0.94) to this set and describes each estimator's sensitivity.

Estimator	Uses	Report description
Close-to-Close (20D)	Closing prices	Standard benchmark
EWMA (lambda 0.94)	Closing returns	Fast regime sensitivity
Parkinson	High and low	Range-efficient
Garman-Klass	Open, high, low, close	Combines opening jumps and range
Yang-Zhang	Overnight gap plus session	Minimum-variance, drift-robust

Capability coverage

Module	Analysis	Modelling	Risk	Visualisation	Export
Return Analytics	Yes	No	Yes	Yes	Yes
Statistical Analysis	Yes	Yes	Yes	Yes	Yes
Volatility Lab	Yes	Yes	Yes	Yes	Yes

5. Forecasting

Three pages forecast the selected asset with classical econometrics, supervised machine learning and recurrent neural networks. All three validate on held-out data in time order, never on shuffled samples.

Module	Page file	Model families
Time Series Forecasting	`pages/15_Time_Series_Forecasting.py`	ARIMA, SARIMA, decomposition
ML Forecasting	`pages/16_ML_Forecasting.py`	Linear models, tree ensembles
DL Forecasting	`pages/17_DL_Forecasting.py`	LSTM, GRU

Time Series Forecasting

- ARIMA and seasonal SARIMA specification, with automatic order selection.
- Multi-step forecast trajectory with 95% confidence bounds that widen with the horizon.
- STL and classical decomposition into trend, seasonal and residual components.
- Model selection by AIC, BIC and HQIC.

In the report, forecasts cover 20 trading days with bounds of plus or minus 1.96 standard errors. Models are benchmarked on AIC, BIC, in-sample RMSE and in-sample MAE. In-sample fit is not out-of-sample accuracy, and the report says so.

The report's decomposition is additive. It uses a centred 20-day moving-average trend and accounts for variance with an explicit cross-covariance term, so the shares sum to 100%.

```
Var(Y) = Var(T) + Var(S) + Var(\varepsilon) + 2\sum Cov
```

ML Forecasting

- Linear, Ridge, Lasso and ElasticNet regressors.
- Random Forest and gradient-boosted decision trees. XGBoost, LightGBM and CatBoost are all installed.
- Chronological 80/20 train-test split.
- Feature embeddings built from lagged returns, rolling moments and technical indicators.
- Feature-importance rankings and out-of-sample directional accuracy.

The report scores each model on test R-squared, MAE, RMSE and training time. Its feature table lists five return lags, 20-day rolling volatility annualised by the square root of 252, and 10-day momentum defined as Close divided by SMA(10), minus 1.

DL Forecasting

- LSTM and GRU sequence networks, with a vanilla RNN named in the report.
- Multi-epoch loss-convergence curves.
- Out-of-sample multi-step predictions.

The report specification for these networks is a 60-step lookback window, 0.20 dropout, MSE loss, the Adam optimiser at learning rate 0.001, and up to 50 epochs with early stopping after 10 epochs without improvement. They run on `tensorflow-cpu` and Keras 3.

Capability coverage

Module	Analysis	Modelling	Visualisation	Export
Time Series Forecasting	No	Yes	Yes	Yes
ML Forecasting	No	Yes	Yes	Yes
DL Forecasting	No	Yes	Yes	Yes

6. Simulation & Risk

Three pages quantify uncertainty: simulated futures, tail-loss measures, and hidden market states.

Module	Page file	Role
Monte Carlo Simulations	`pages/14_Monte_Carlo_Simulations.py`	Multi-path price simulation (featured workspace)
Risk Analytics	`pages/13_Risk_Analytics.py`	VaR, expected shortfall, drawdowns (featured workspace)
Regime Detection	`pages/06_Regime_Detection.py`	Hidden Markov market states

Monte Carlo Simulations

- Three path generators: Geometric Brownian Motion, Merton Jump Diffusion and bootstrap resampling.
- 500 or more paths, drawn as a quantile fan chart with the 5th, 25th, 50th, 75th and 95th percentiles.
- Terminal price distribution and the probability of loss.

The report uses 500 GBM iterations over a 60-day horizon and quotes the probability of loss and the median terminal price.

Risk Analytics

- Value at Risk by three methods: historical, parametric and Cornish-Fisher, each at 95% and 99%.
- Conditional VaR (expected shortfall).
- Underwater drawdown depth and recovery duration.
- Historical stress tests and left-tail loss CDF curves.

The report defines its metrics as follows.

Metric	Definition
Historical VaR (95%, daily)	5th percentile of daily returns
Parametric VaR (95%, daily)	Mean minus 1.645 times the standard deviation
CVaR / expected shortfall	Mean return given a return below VaR
Tail ratio	Average right tail (95th) divided by left tail (5th)
Downside deviation	Dispersion of negative returns, annualised by the square root of 252
Sortino ratio	(Annualised return minus risk-free rate) divided by downside deviation
Calmar ratio	CAGR divided by the absolute maximum drawdown

The report page defaults the risk-free rate to 6.5% a year for India and 4.5% for the US, and the user can change it.

Regime Detection

- Gaussian Hidden Markov Models with 2 or 3 states.
- Regime transition probability matrices.
- State-conditional mean return and volatility.
- Smoothed state-probability time series.

The report describes the model as three latent states fitted by Baum-Welch expectation-maximisation and decoded with the Viterbi algorithm, using a uniform initial prior and a row-stochastic transition matrix. Section 9 explains why this report page can show "not executed".

Capability coverage

Module	Analysis	Modelling	Simulation	Risk	Visualisation	Export
Monte Carlo Simulations	No	Yes	Yes	Yes	Yes	Yes
Risk Analytics	Yes	No	No	Yes	Yes	Yes
Regime Detection	Yes	Yes	No	Yes	Yes	Yes

7. Strategy & Trading

Three pages cover systematic strategy research, a full backtesting terminal, and a reinforcement-learning sandbox. Strategy Lab is for exploring and comparing ideas; Backtesting is for validating one strategy in depth.

Module	Page file	Role
Strategy Lab	`pages/08_Strategy_Lab.py`	Strategy catalogue, signal visualiser, comparison
Backtesting	`pages/10_Backtesting.py`	Four-tab backtest terminal (featured workspace)
RL Trading	`pages/18_RL_Trading.py`	Reinforcement-learning agent environment

Strategy Lab

A single-page workstation for building, validating and stress-testing systematic strategies. Every strategy emits a signal series where +1 is a buy entry, -1 is an exit and 0 is no action. Signals are generated without look-ahead.

Family	Strategy	Entry (+1)	Exit (-1)
Technical	Moving Average Crossover	Fast MA crosses above slow MA (SMA or EMA, default 20 and 50)	Fast MA crosses below slow MA
Technical	RSI Mean Reversion	RSI crosses back above oversold (period 14, 30 and 70)	RSI crosses above overbought
Technical	Bollinger Bands Breakout / Squeeze	Breakout mode: close crosses above the upper band. Squeeze mode: close crosses below the lower band	Breakout: close falls below the middle band. Squeeze: close crosses above the upper band
Technical	MACD Trend Momentum	MACD crosses above signal (12, 26, 9)	MACD crosses below signal

Family	Strategy	Entry (+1)	Exit (-1)
Technical	Donchian Channel Breakout	Close above the prior 20-bar high	Close below the prior 10-bar low
Technical	Momentum / Rate of Change	ROC (20 bars) crosses above the threshold	ROC crosses below zero
Statistical	Z-Score Mean Reversion	Z-score crosses below minus the entry level (20 bars, entry 2.0)	Z-score crosses above the exit level (0.0)
Statistical	Pairs Trading / Spread Reversion	Price-ratio z-score crosses below minus entry (30 bars, entry 2.0) against a second symbol	Z-score crosses above minus exit (0.5)
Factor	Momentum Factor (12M minus 1M, volatility-adjusted)	Score crosses above 0.5	Score crosses below -0.2
Factor	Value / Long-Term Mean Reversion	Price falls below its 200-day median by 10%	Price rises above the median by 5%
Factor	Low Volatility Anomaly	21-day volatility crosses below its 126-day average while price is above SMA 50	Volatility exceeds 1.3 times its average, or price falls below SMA 50
Custom	Custom Strategy Builder	Two conditions joined by AND or OR	One exit condition

The custom builder compares one of four indicators (RSI 14, close versus SMA 200, close versus SMA 50, SMA 20 versus SMA 50) with a threshold using `<`, `<=`, `>`, `>=` or `==`.

Other capabilities of the page:

- **Universe loader:** reads the India and US snapshot CSVs (price, market cap, P/E, EPS, EPS growth, dividend yield, volume, sector) and caches them for 24 hours.
- **Market data:** Daily, Weekly or Monthly bars. Data is rejected if it is empty, has fewer than 30 bars, contains non-positive prices, or is missing more than 10% of closes.

- **Strategy visualiser:** for each strategy, shows its formula, the regime it suits, its main risk, a typical holding horizon, today's signal state, and a price and indicator chart with buy and sell markers.
- **Instant preview:** a quick backtest on 1,000,000 of capital reporting CAGR, total return, Sharpe, Sortino, maximum drawdown, longest drawdown in days, win rate and profit factor.
- **Comparison and tuning (per module registry and README):** composite logic builder (AND, OR, majority vote), tournament leaderboard, 2D hyperparameter heatmaps, and walk-forward validation.

Backtesting

An institutional backtest terminal organised into four tabs.

Tab	Contents
Overview	KPI strip, strategy against benchmark, key metrics, trade distribution donut, monthly heatmap, drawdown episodes, recent trades, settings
Performance Analysis	Equity curve with buy and sell markers, drawdown duration and recovery, rolling 252-day metrics, risk-return scatter, yearly performance, statistics table
Trade Analysis	Eight trade KPIs, per-trade PnL, cumulative PnL, duration and return histograms, outcome donut, entry and exit reasons, full trade log, trade inspector
Strategy Comparison	Multi-select strategy cards, cumulative returns, risk-return scatter, side-by-side metrics matrix, drawdown chart

Controls and defaults:

Control	Options	Default
Exchange	NSE, BSE, NASDAQ, NYSE	Follows sidebar region

Control	Options	Default
Strategy	SMA Crossover, EMA Crossover, RSI, MACD, Bollinger Bands, Donchian Breakout, Momentum, Mean Reversion (Z-Score), Breakout, Buy & Hold	SMA Crossover (20, 50)
Benchmark	NIFTY 50, SENSEX, Bank Nifty, S&P 500, NASDAQ 100, Buy & Hold Asset	NIFTY 50 (India) or S&P 500 (US)
Capital	1,000 or more	100,000
Horizon	1Y, 3Y, 5Y, 10Y, Max, Custom dates	5Y
Position mode	Long Only, Long & Short	Long Only
Brokerage	0 to 1%	0.05%
Slippage	0 to 1%	0.01%
Stop loss, take profit	0 to 50% and 0 to 100%, where 0 disables	Disabled

How the simulation works:

- The engine steps through the data bar by bar. A signal read at the close of bar i-1 is acted on at the open of bar i, so there is no look-ahead.
- Stop-loss and take-profit levels are checked against each bar's low and high. If a bar gaps through a level, the exit fills at the open, not at the level.
- Exits also occur on a signal reversal. Each trade pays the commission rate on both entry and exit notional.
- The trade log records trade number, side, entry and exit dates and prices, quantity, gross PnL, fees, net PnL, return percent, duration in days, entry and exit reasons, and win or loss status.
- Price history comes from a dual-engine fetcher: `yf.Ticker.history` first, then `yf.download`, retrying shorter periods if `max`, `5y` or `3y` fails. The benchmark is aligned to the asset's trading days.
- A symbol map handles renamed or special-character tickers. It maps `TATAMOTORS` to `TMPV`, and turns underscores into `&` or `-` (for example `M_M` to `M&M`).

RL Trading

- A Gymnasium trading environment with discrete and continuous action spaces.
- Q-Learning and policy-gradient agents.
- Custom risk-adjusted reward functions.
- Action trajectories and a policy performance audit.

This is the only page that does not support export.

Capability coverage

Module	Modelling	Simulation	Backtesting	Risk	Export
Strategy Lab	Yes	No	Yes	Yes	Yes
Backtesting	No	No	Yes	Yes	Yes
RL Trading	Yes	Yes	Yes	No	No

8. Portfolio & Research

Three pages work across assets: allocation, factor exposure and pairs trading. The first two feed the correlation and factor pages of the PDF report.

Module	Page file	Role
Portfolio Lab	`pages/11_Portfolio_Lab.py`	Mean-variance and risk-parity allocation
Factor Research	`pages/07_Factor_Research.py`	Beta, alpha and factor attribution
Statistical Arbitrage	`pages/09_Statistical_Arbitrage.py`	Cointegration and pairs trading

Portfolio Lab

- Markowitz efficient frontier generation.
- Maximum Sharpe and minimum variance portfolios.
- Equal Risk Contribution (ERC) and inverse-volatility weighting.
- Black-Litterman allocation with subjective views and covariance shrinkage.

Optimisation uses `cvxpy` and `PyPortfolioOpt`.

Factor Research

- CAPM market beta and Jensen's alpha.
- Fama-French style sensitivity regressions.
- Rolling systematic beta over time.
- Share of variance explained by the benchmark (R-squared) and residual fit.

The shared metrics library defines beta and alpha on per-period returns, with no annualisation.

```
\beta = \frac{Cov(R_p, R_b)}{Var(R_b)}, \qquad \alpha = \bar{R}_p - \left[
r_f + \beta\,(\bar{R}_b - r_f) \right]
```

Statistical Arbitrage

- Engle-Granger two-step cointegration test.
- Pairs spread construction with a dynamic hedge ratio.
- Ornstein-Uhlenbeck mean-reversion half-life estimate.
- Z-score threshold signals and a pair backtest.

The half-life is the time for a spread to close half the distance back to its mean. A short half-life suggests a faster-reverting, more tradable pair.

The PDF report rates the portfolio, factor and stat-arb pages as partial, because they need more than one ticker configured.

Capability coverage

Module	Analysis	Modelling	Backtesting	Risk	Export
Portfolio Lab	No	Yes	No	Yes	Yes
Factor Research	Yes	Yes	No	Yes	Yes
Statistical Arbitrage	Yes	Yes	Yes	Yes	Yes

9. Research Reporting

The Report Generation Center turns the current session into a 20-page institutional PDF. It refuses to run without valid market data, and any module that has not been run appears as a specification, never as invented numbers.

Component	File	Role
Page	`pages/19_Report_Generation.py`	Validation, configuration, readiness panel, download
Data assembly	`reporting/report_data.py`	`build_report_data` and `validate_report_data`
PDF engine	`reporting/pdf_generator.py`	`generate_pdf_report`, which returns the PDF as an in-memory buffer
Styling	`reporting/report_styles.py`	Fonts, page geometry, numbered canvas, callout boxes
Charts	`reporting/report_charts.py`	One chart function per report panel

Workflow

1. The page loads data for the sidebar ticker, period and interval, and removes non-trading rows.
2. `validate_report_data` checks the data and shows ticker, daily rows, columns, start and end dates and latest close. If there are no observations, generation stops.
3. The user sets the report options (below).
4. `build_report_data` computes the cross-module analytics. It also picks up Monte Carlo results saved in session state under `mc_sim_results`.
5. `generate_pdf_report` builds the PDF with ReportLab.
6. The page offers a download named `QuantTerminal_Report_<TICKER>_<YYYYMMDD>.pdf`.

Report options

Option	Choices	Default
Target ticker	Taken from the sidebar, read-only	Sidebar ticker

Option	Choices	Default
Benchmark	NIFTY 50, S&P 500, NASDAQ 100, BSE SENSEX	NIFTY 50 (India) or S&P 500 (US)
Report type	Complete Quantitative Report (20 pages), Executive Risk Brief, Factor & Arbitrage Memo	Complete report
Risk-free rate	0% to 20% annualised	6.5% (India), 4.5% (US)
Included modules	15 checkboxes, from Market & Price Dynamics to Factor Research & Stat-Arb	All selected

Page-by-page contents

Page	Section	Contents
1	Executive Analytics Summary	Six KPI cards (latest price, CAGR, volatility, maximum drawdown, historical VaR, 60-day probability of loss), readiness chart, domain coverage table, observations
2	Data Provenance and Methodology	Observations and window, pipeline timeline, provenance table, methodology table, reproducibility statement
3	Market Dynamics	Price and volume chart with moving averages, return profile table
4	Technical Indicators	SMA 20, 50 and 200, RSI 14, MACD, Bollinger upper band, ATR 14
5	Return Moments and Tests	Distribution histogram, KDE and Q-Q plot, mean, volatility, skewness, kurtosis, Jarque-Bera, Shapiro-Wilk
6	Volatility Dynamics	Five estimators plus EWMA, with formula and sensitivity
7	Tail Risk and Drawdowns	Historical and parametric VaR, CVaR, tail ratio, downside deviation, Sortino, Calmar

Page	Section	Contents
8	Regime Detection	Regime timeline, or the HMM specification if not run
9	Time Series Decomposition	Trend, seasonal and residual variance shares
10	Model Benchmark	ARIMA specifications compared on AIC, BIC, RMSE and MAE
11	Forecast Horizon	20-day forecast cone with T+1, T+5 and T+20 targets
12	Machine Learning	Regressor comparison on an 80/20 split
13	Deep Learning	LSTM, GRU and RNN specification, with no metrics unless trained
14	Strategy Execution	Systematic strategy execution and cumulative equity
15	Strategy Comparison	Multi-strategy comparative benchmark
16	Price Path Simulation	Stochastic price path simulation and fan profiles
17	Terminal Distribution	Terminal price distribution and tail-loss probabilities
18	Correlation and Portfolio	Multi-asset correlation matrix and portfolio allocation
19	Factors and Pairs	Factor beta sensitivities and cointegration pairs
20	Synthesis and Governance	Research synthesis, limitations and disclaimers

Layout and typography

- US Letter pages, with a target of 85% to 95% of the printable area filled.
- DejaVu Sans, so the rupee and dollar symbols render without replacement boxes.
- Reusable blocks: a KPI card row, standard tables with a coloured header and zebra rows, a blue observations box, and a callout box.
- Every page starts with a section tag, a title and a subtitle.

Integrity behaviour

- When a module has not been run, its page shows a specification table with a status such as Not Executed, Not Evaluated or Pending, a placeholder chart, and a callout saying no result is displayed.
- Number formatting is safe: missing, NaN or infinite values print as a dash.

10. Data Layer, Shared Controls and Core Metrics

Every page reads the same ticker, period and interval from a shared sidebar, loads prices through one cached function, and cleans them the same way. This section documents that shared layer.

Market coverage

Region	Exchanges	Yahoo suffix	Currency	Cap bands (as implemented)
India	NSE, BSE	`.NS`, `.BO`	INR	Large 75,000 Cr or more; Mid 20,000 Cr or more; Small 1,000 Cr or more; Micro below
US	NASDAQ, NYSE	None	USD	Large 10B or more; Mid 2B or more; Small 300M or more; Micro below

These bands are in `categorize_market_cap` in `utils/helper.py`. A missing or non-positive market cap is treated as Micro Cap.

Data sources

- **Prices:** Yahoo Finance daily OHLCV through `yfinance`, adjusted for splits and dividends. Company info and financial statements are fetched separately and cached for one hour.
- **Universe lists:** `data/snapshots/India_Stocks_Data.csv` and `US_Stocks_Data.csv`, falling back to `data/` if the snapshot folder is missing. Columns include symbol, description, ISIN, exchange, sector and market capitalisation. Header-like rows are filtered out so they never reach Yahoo.

Loading and hygiene (`utils/helper.py`)

- **`load_data(ticker, period, interval)`** downloads with auto-adjustment. If a 5-year request comes back empty (a known Yahoo glitch on `.NS` tickers), it downloads the maximum history and keeps the last 1,260 bars. If the result is still empty, it retries without auto-adjustment. It flattens multi-index columns and returns an empty frame on any failure.
- **`drop_holiday_nans(df)`** removes rows where all price columns are missing, and rows where volume is zero or missing and open, high, low and close are all equal. These flat, zero-volume rows are how holiday placeholders appear in the feed. Removing them stops spurious zero returns from distorting autocorrelation and signals.
- **Caching:** `st.cache_data` wraps data loads. The Backtesting page uses its own 30-minute cache and a Clear Cache control.

Periods map to the intervals Yahoo allows:

Period	Allowed intervals
1d, 5d	1m, 2m, 5m, 15m, 30m, 60m, 90m
1mo	30m, 60m, 90m, 1d
3mo, 6mo, 1y, 2y, 5y, 10y, max	1d, 5d, 1wk, 1mo

Shared sidebar (`utils/sidebar.py`)

`render_sidebar()` draws the same controls on every page and returns `(ticker, company, exchange, period, interval, region)`.

Control	Behaviour
Market Region	India or US
Exchange	Options come from the snapshot CSV (for example NSE and BSE)
Market Cap	All, Large, Mid, Small or Micro. If a band is empty, the sidebar warns and shows all stocks for that exchange

Control	Behaviour
Stock / Company	Searchable list sorted by company name, with a count of available stocks
Yahoo Finance Ticker	Read-only preview. India adds `.NS` or `.BO`; US uses the plain symbol
Period, Interval	Interval choices depend on the period

The sidebar stores its choices in session state so pages stay in sync: `region`, `exchange`, `market_cap_filter`, `stock_option`, `ticker`, `selected_ticker`, `company`, `symbol`, `isin`, `period`, `selected_period`, `interval` and `selected_interval`.

Configuration (`config.py`)

Constant	Value
Exchange options	Auto, NSE, BSE, GLOBAL
India benchmarks	Nifty 50 (`^NSEI`), Sensex (`^BSESN`), Bank Nifty (`^NSEBANK`)
Global benchmarks	S&P 500 (`^GSPC`), NASDAQ (`^IXIC`), None
Currency symbols	INR, USD, EUR, GBP, JPY, SGD
Indian presets	20 large-cap NSE tickers, from RELIANCE.NS to ONGC.NS
Default risk-free rate	5% (0.05)
Trading days per year	252

Core metrics library (`core/metrics.py`)

The test suite pins these definitions. Inputs must be a single return series, not a table.

Metric	Definition	Annualised
Sharpe	(Mean daily return times 252, minus risk-free rate) divided by (daily standard deviation times the square root of 252)	Yes

Metric	Definition	Annualised
Sortino	As Sharpe, but the denominator uses only the standard deviation of negative returns	Yes
Calmar	CAGR divided by the absolute maximum drawdown, where CAGR = (end / start) raised to 1 / (bars / 252), minus 1	Yes
Information ratio	Mean of active return divided by its standard deviation, where active return is portfolio minus benchmark	No
Treynor	(Mean return minus risk-free rate) divided by beta	No
Beta	Covariance with benchmark divided by benchmark variance	No
Alpha	Mean return minus [risk-free + beta times (benchmark mean minus risk-free)]	No

Edge cases: Sharpe of a constant series is infinite; Sortino with no negative returns, Calmar with no growth or drawdown, and any ratio against a constant benchmark return NaN. Information ratio, Treynor, beta and alpha raise an error when no benchmark is given. Drawdown comes from `core/drawdown.py`.

Theme and formatting

`inject_custom_theme()` applies the dark glass-style theme (Inter and JetBrains Mono fonts, styled metric cards, tables, tabs and buttons) to each page. Formatting helpers produce compact numbers (K, M, B, T), currency strings and percentages, and show a dash for missing values.

11. Testing and Verification

The test suite uses pytest. The metrics tests check each function against its documented formula, not just that it returns a value.

```
pytest                                    # all tests
pytest tests/test_metrics.py              # core risk and performance metrics
pytest tests/test_reporting_pipeline.py   # 20-page PDF pipeline
```

`tests/test_metrics.py`

33 tests in eight classes cover `core/metrics.py`.

Class	Tests	What is checked
TestSharpeRatio	6	Documented formula, 252-day annualisation, risk-free rate, constant series gives infinity, a hand-computed value of 22.4499 for returns of 1% and 3%
TestSortinoRatio	4	Formula, downside-only denominator, risk-free rate, all-positive returns give NaN
TestCalmarRatio	4	Formula, use of the shared drawdown function, no risk-free parameter, flat equity gives NaN
TestInformationRatio	4	Formula, no annualisation, benchmark required, identical series give NaN
TestBeta	4	Covariance over variance, benchmark required, constant benchmark gives NaN, scaling behaviour
TestTreynorRatio	3	Formula, no annualisation, benchmark required
TestAlpha	5	CAPM expression, risk-free rate, no annualisation, benchmark required, zero alpha when portfolio equals benchmark
TestSeriesOnlyContract	3	Sharpe, Sortino and Calmar reject a multi-column table

`tests/test_reporting_pipeline.py`

Per the README, this file tests the 20-page PDF report pipeline.

Runtime safeguards outside the test suite

- The report page validates market data before any computation and stops on empty or invalid data.
- Strategy Lab rejects datasets with fewer than 30 bars, non-positive prices, or more than 10% missing closes.

- The Backtesting page stops with an error and a Clear Cache button when fewer than 25 bars load.

12. Limitations and Governance

QuantTerminal is for education and research. Its outputs are conditional historical estimates, not investment, tax or financial advice, and the README states this on every deployment.

Limitations

- **Data source:** prices come from Yahoo Finance through an unofficial library. The code carries workarounds for empty 5-year downloads, so availability and history depth can vary by ticker.
- **Single-asset focus:** most pages analyse one ticker at a time. Portfolio, factor and pairs analysis need more than one ticker configured.
- **In-sample versus out-of-sample:** ARIMA benchmarking uses in-sample fit, which the report says does not measure forecast accuracy.
- **Heavy models:** deep-learning models need training time, and the report treats them as not run until trained.
- **Backtests:** results depend on the chosen costs and horizon. They describe the past and do not predict returns.